

Aaron Chou

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EDUCATION

Carnegie Mellon University (CMU) , <i>M.S. in Robotic Systems Development (MRSD)</i>	Pittsburgh, PA
• Coursework: Introduction to Robot Learning	Aug. 2026 – May 2028
National Taiwan University (NTU) , <i>B.S. in Mechanical Engineering</i>	Taipei, Taiwan
• Last 60 GPA: 4.03/4.3 CGPA: 3.80/4.3	Sept. 2018 – Jan. 2023
• Coursework: Automatic Control, Digital Control System, Kinematics, Dynamics, Computer Programming	
Aoyama Gakuin University (AGU) , <i>Exchange Program</i>	Tokyo, Japan, Fall 2022

PUBLICATIONS

- [1] **Y.L. Chou**, L.C. Wang, Y.S. Luo, Y.T. Liu, J.L. Li. *VL-N3RD-Bench: Benchmarking Vision-Language Navigation with 3D Gaussian Splatting Reconstruction for Deployment*, ICRA Workshop: VLA Pipelines for Real Robots, 2026. ([link](#))
- [2] Y.S. Luo, L.C. Wang, H. Mandala, **Y.L. Chou**, G.H.G. Christmann, C.Y. Lee, W.C. Chen, et al. *Feasibility-Guided Planning over Multi-Specialized Locomotion Policies*, IEEE Int. Conf. Robot. Autom. (ICRA), 2026. ([link](#))

EXPERIENCE

Stealth-Mode Startup	Mountain View, CA
<i>Robotics Research Intern — UMI, Diffusion Policy</i>	May 2026 – Aug. 2026
Inventec Corporation , AI Center, Robotics	Taipei, Taiwan
<i>Robotics Research Engineer — Quadruped Robots</i>	Mar. 2025 – May 2026
• Vision-Language Navigation (VLN): Integrated and evaluated a pretrained VLA model for real-world navigation, and built an IsaacLab benchmark to analyze how 3D Gaussian Splatting (3DGS) reconstruction quality affects navigation performance.	
• 3DGS Benchmarking: Benchmarked SoTA 3DGS pipelines using standard image-quality metrics and runtime/memory analysis.	
• Learning-Based Control: Developed low-level control infrastructure with LCM communication, custom motor interfaces, calibration routines, and IMU integration for quadruped platforms with no official API, deploying IsaacGym -trained policies (link).	
• Terrain-Aware Navigation: Trained a 15 cm gap-traversal policy, implemented a feasibility-guided planning framework for hybrid terrain skill selection, conducted real-world experiments, and contributed to an ICRA 2026 submission [2].	
• SLAM Optimization: Improved single-LiDAR SLAM by dual-LiDAR fusion (link) in ICRA 2025 QRC, achieving a 65% reduction in mapping-to-navigation time and enabling rapid terrain reconstruction in dynamic environments.	
Flytech Technology Co. Ltd.	Taipei, Taiwan
<i>Robotics Engineer — Autonomous Mobile Robots</i>	Mar. 2024 – Jan. 2025
• Engineered a precise docking system (demo) by integrating motion control, localization, and a customized rack tracker, achieving navigation accuracy of $\pm 1\text{--}1.5\text{ cm}$ to minimize redocking attempts in production deployments.	
• Developed and maintained 10+ ROS packages and delivered a patrol demo showcased at Computex 2024.	
Chien Kuo High School , Robotics team, FRC#8020 CyberpunkK	Taipei, Taiwan
<i>Youth Mentor — Mechanical Design</i>	Feb. 2022 – Jul. 2023
• Mentored 30+ students for robot design, CAD/CAM, CNC operation, and testing of the competition-ready robot (demo).	

SKILLS

Robotics: IsaacSim/Lab, ROS/ROS2 (Nav2, MoveIt2, FAST-LIO), 3D Reconstruction (3DGS), Sensor Fusion, SLAM

Hardware & Software: Unitree Go1/A1, Flexiv Rizon, Livox LiDAR, RealSense, IMU, RTX 5090; Docker, Git, Blender, OnShape

Programming: Python, C++, Shell Scripting (Bash), LaTeX

HONORS & COMPETITIONS

2025 ICRA Quadruped Robot Challenge (QRC) — Participation Certificate, Autonomous (demo)	Atlanta, GA
2022 FIRST Robotics Competition (FRC) Sacramento Regional, Finalist (Team Mentor)	Sacramento, CA
2020 Dean's List Award (top 5% of the class in the semester)	Taipei, Taiwan